

4(a)	<table border="1"> <thead> <tr> <th data-bbox="336 215 1220 311">statement</th><th data-bbox="1220 215 1545 311">genetic engineering using a plasmid</th><th data-bbox="1545 215 1870 311">gene editing</th></tr> </thead> <tbody> <tr> <td data-bbox="336 311 1220 375">It can add a new phenotypic characteristic to an organism.</td><td data-bbox="1220 311 1545 375">✓</td><td data-bbox="1545 311 1870 375">✓ ;</td></tr> <tr> <td data-bbox="336 375 1220 438">It can change an A-T base pair to C-G.</td><td data-bbox="1220 375 1545 438"></td><td data-bbox="1545 375 1870 438">✓ ;</td></tr> <tr> <td data-bbox="336 438 1220 502">It can inactivate a desired selected gene in an organism.</td><td data-bbox="1220 438 1545 502"></td><td data-bbox="1545 438 1870 502">✓ ;</td></tr> <tr> <td data-bbox="336 502 1220 606">It may change DNA in a way that cannot be told apart from a natural mutation.</td><td data-bbox="1220 502 1545 606"></td><td data-bbox="1545 502 1870 606">✓ ;</td></tr> <tr> <td data-bbox="336 606 1220 678">It requires a DNA donor and a recipient.</td><td data-bbox="1220 606 1545 678">✓</td><td data-bbox="1545 606 1870 678">;</td></tr> </tbody> </table>	statement	genetic engineering using a plasmid	gene editing	It can add a new phenotypic characteristic to an organism.	✓	✓ ;	It can change an A-T base pair to C-G.		✓ ;	It can inactivate a desired selected gene in an organism.		✓ ;	It may change DNA in a way that cannot be told apart from a natural mutation.		✓ ;	It requires a DNA donor and a recipient.	✓	;	5
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4(b)(ii)	<i>any three from:</i> 1 less food waste ; A increases quality of food 2 <i>idea of</i> less food shortages / more food production / helps solve global demand for food ; 3 more income for, growers / farmers or ref. economic benefit for, country / region ; 4 crop can be grown, when there is a water shortage / in poor quality soil / in harsh environment ;	3																		